

WEEK – 2 DOCUMENTATION

Initial Setup commands :-

#Create an virtual environment and activate it.

python -m venv env

source env/bin/activate

pip install -r requirements.txt

pip install "dvc[gs]"

#clone the repository, if its a private repo follow these instructions

ssh-keygen -t ed25519 -C "vertex-workbench(optional)"

To view the public key which should be provided to github

cat ~/.ssh/id_ed25519.pub

TASK – 1

#Initialize DVC

dvc init

#Verify DVC initialization

ls -la

#Push it to git

git add .

git commit -m "Intialize DVC"

git push origin week_2

TASK – 2

#To check whether the shell has permission to access cloud storage (Task - 2)

gcloud auth login

gcloud auth application-default login

To revoke authentication

gcloud auth application-default revoke

To list which account is active

gcloud auth list

#To set remote path

dvc remote add -d myremote gs://your-bucket/week_2/dvc

#To modify remote path

dvc remote modify myremote url gs://your-bucket/week_2/dvc

#Verify

dvc remote list

#Commit DVC config

git add .

git commit -m "Configure GCS remote for DVC"

git push origin week_2

TASK – 3

#Get file using wget (Iteration - 1)

wget https://raw.githubusercontent.com/IITMBSMLOps/ga_resources/week_1/data/raw/iris.csv

#Run python script

python train.py

#Track with DVC

dvc add iris.csv model.joblib

#Update git

git add .

git commit -m "Iteration 1"

#Push to GCS

dvc push

#To check the files that are tracked by git

git ls-files

#Get file using wget and overwrite the existing IRIS dataset(Iteration - 2)

**wget https://raw.githubusercontent.com/IITMBSMLOps/ga_resources/week_1/data/v1/data.csv
-O iris.csv**

#Run python script

python train.py

#Track with DVC

dvc add iris.csv model.joblib

#Update git

git add .

git commit -m "Iteration 2"

#Push to GCS

dvc push

#Get file using wget and overwrite the existing IRIS dataset(Iteration - 3)

wget https://raw.githubusercontent.com/IITMBSMLOps/ga_resources/week_1/data/v2/data.csv
-O iris.csv

#Run python script

python train.py

#Track with DVC

dvc add iris.csv model.joblib

#Update git

git add .

git commit -m "Iteration 3"

#Push to GCS

dvc push

#If everything is synchronized, DVC should indicate that the remote is up to date.

dvc status -c

#updating the repository

git push origin week_2

TASK – 4

#Switching between versions

#Count the the rows of the dataset to note changes

wc -l iris.csv

#list the model with timestamp to note changes

ls -lh model.joblib

#To list the logs

git log --oneline

#Iteration - 1

#Switch to old version/commit

git checkout <hash_of_respective_commit>

#update the dataset and models from cache

dvc checkout

#Count the the rows of the dataset to note changes
wc -l iris.csv

#list the model with timestamp to note changes
ls -lh model.joblib

#Iteration - 2
#Switch to old version/commit
git checkout <hash_of_respective_commit>

#update the dataset and models from cache
dvc checkout

#Count the the rows of the dataset to note changes
wc -l iris.csv

#list the model with timestamp to note changes
ls -lh model.joblib

#Iteration - 3
#Switch to old version/commit
git checkout <hash_of_respective_commit>

#update the dataset and models from cache
dvc checkout

#Count the the rows of the dataset to note changes
wc -l iris.csv

#list the model with timestamp to note changes
ls -lh model.joblib

#Finally update the repo with the latest commit
git checkout week_2